



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - I - Apr - 2026

II Semester

22MA201 & ENGINEERING MATHEMATICS II

1	(i)	A robot moves in straight lines described by $y = mx + c$ where slope varies with terrain. Form differential equation when both m and c are arbitrary and justify order. (3 Marks - [U/C, 2])
2	(ii)	A sensor output follows $y = e^{bx}(a \cos x)$ Identify the order of resulting differential equation and justify. (2 Marks - [U/C, 1])
3	(i)	A body is dropped from rest. Formulate the differential equation and find velocity. (3 Marks - [U/C, 2])
4	(ii)	Apply Newton's law of cooling to formulate the differential equation for an object at $90^\circ C$ placed in a surrounding of $25^\circ C$. (2 Marks - [U/C, 1])
5	(i)	Fluid flow velocity is given by $y = \frac{a}{x} + b$ where constants depend on system pressure. Form the differential equation and interpret the result. (3 Marks - [U/C, 2])
6	(ii)	A sensor output follows $y = e^x(a \cos x + b \sin x)$. Identify the order of resulting differential equation and justify. (2 Marks - [U/C, 1])
7	(i)	A sensor cools from $90^\circ C$ to $70^\circ C$ in 8 minutes in a $20^\circ C$ room. Find the cooling constant k . (3 Marks - [U/C, 2])
8	(ii)	Apply the concept of free fall to construct the differential equation and determine the velocity of a body dropped from rest. (2 Marks - [U/C, 1])
9	(i)	In an electric vehicle battery cooling system, the temperature T satisfies $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. Solve using separation of variables and evaluate the time required for the battery to reach a safe temperature given initial conditions. (5 Marks - [Ap/C, 3])

10	(i)	A vehicle slows down in water where deceleration is proportional to velocity. Initial velocity is 20 m/s. After 5 seconds, velocity becomes 10 m/s. Formulate, solve, and interpret velocity after 10 seconds. (5 Marks - [An/C, 2])
11	(i)	In a wireless sensor system, signal strength S decays as $\frac{dh}{dt} = -k\sqrt{h}$ Solve the equation using separation of variables and analyze how initial signal strength influences transmission range. (5 Marks - [An/C, 3])
12	(i)	A bacterial culture grows from 500 to 1500 in 4 hours. Develop the model and predict population after 6 hours. (5 Marks - [Ap/P, 3])
13	(i)	A damping system in machinery follows $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$ An engineer incorrectly assumes integrating factor e^{-2x}, leading to wrong results. Identify the mistake, compute correct integrating factor, and solve the equation. (5 Marks - [An/C, 3])
14	(i)	A viral infection spreads in a closed campus of 5000 students. Initially, 5 students are infected and after 4 days, 50 students are infected. Formulate the logistic model, determine constants, and evaluate infected population after 10 days. (5 Marks - [An/P, 3])
15	(i)	In an electric vehicle, the battery pack experiences non-uniform heating during fast charging. The temperature rise is influenced by internal resistance and external cooling, leading to the model $\frac{dy}{dx} + 3y = 2x - 1$ where y represents temperature deviation and x represents time. Find the temperature distribution using Leibnitz method and analyze whether the system reaches thermal stability. (5 Marks - [An/C, 3])
16	(i)	A chemical reactor produces a substance limited by capacity 1000 units. Initially 50 units exist and after 2 hours it becomes 200 units. Formulate the logistic equation and analyze production after 6 hours. (5 Marks - [Ap/C, 3])
17	(i)	Classify and justify the equation: $d^2y/dx^2 + 3y = \sin x$ (3 Marks - [U/C, 2])
18	(ii)	Interpret the role of particular integral in second order systems in physical and engineering contexts. (2 Marks - [U/C, 1])

19	(i)	A mass-spring system satisfies $x'' + 25x = 0$ Analyze and interpret amplitude, frequency, and nature of motion. (3 Marks - [U/C, 2])
20	(ii)	A mechanical oscillator completes 50 cycles in 10 seconds. Determine its frequency and interpret its engineering significance. (2 Marks - [U/C, 1])
21	(i)	A system follows $d^2y/dx^2 + 4y = 0$, with $y(0) = 2$, $y'(0) = -4$. The complementary function is $y = C_1 \cos 2x + C_2 \sin 2x$. Determine the constants and interpret system behavior. (3 Marks - [U/C, 2])
22	(ii)	Explain the role of initial conditions in second order systems. (2 Marks - [U/C, 1])
23	(i)	An LC circuit has equation $d^2q/dt^2 + (k/m)q = 0$ Evaluate the effect of increasing k on frequency. (3 Marks - [U/C, 1])
24	(ii)	A spring-mass system in vibration analysis is modeled by a second-order differential equation is $\frac{d^2x}{dt^2} + \omega^2 x = 0$. Interpret the type of motion. (2 Marks - [U/C, 1])
25	(i)	An RLC circuit satisfies $d^2i/dt^2 + 4di/dt + 4i = 0$ Determine complementary function and evaluate system response. (5 Marks - [Ap/P, 3])
26	(i)	A vibration system satisfies $d^2x/dt^2 + 8dx/dt + 16x = 0$ Analyze the damping condition and evaluate system response. (5 Marks - [Ap/P, 3])
27	(i)	A spring-mass system is governed by $d^2y/dt^2 + 5dy/dt + 6y = 0$ Formulate the auxiliary equation, determine the complementary function, and analyze system behavior. (5 Marks - [Ap/P, 3])
28	(i)	A vehicle suspension system is modeled by $m d^2x/dt^2 + c dx/dt + kx = 0$ with $m = 2$ kg, $c = 6$ Ns/m, $k = 5$ N/m. Formulate the differential equation, determine the nature of damping, and interpret system response. (5 Marks - [Ap/P, 3])

29	(i)	A vibration system is governed by $\frac{d^2y}{dt^2} + 4y = t^2$ Determine particular integral using undetermined coefficients and evaluate complete response. (5 Marks - [Ap/P, 3])
30	(i)	A structure vibrates under external force $F(t) = 8\cos\omega t$. Analyze and determine the condition for resonance and evaluate its effect. (5 Marks - [An/C, 3])
31	(i)	A control system is described by $\frac{d^2y}{dx^2} + y = \cos 2x$ Determine the particular integral using another method. (5 Marks - [Ap/P, 3])
32	(i)	A spring-mass-damper system is subjected to an external force $F(t) = 10\cos 2t$. Given $m = 1$ kg, $c = 4$ Ns/m, $k = 5$ N/m, formulate the differential equation, determine steady-state solution, and interpret system behavior. (5 Marks - [Ap/P, 3])
33	(i)	An engineer models position as a scalar $x(t) = t^2$ instead of a vector. Identify the error and correct it. (3 Marks - [E/C, 2])
34	(ii)	Temperature is given by $T(x, y) = 5x + 3y$. Identify the type of field. (2 Marks - [E/C, 1])
35	(i)	In a conveyor system, path $\vec{r}(t) = t\hat{i} + t^2\hat{j}$. Identify if motion is straight or curved using tangent direction. (3 Marks - [Ap/P, 2])
36	(ii)	A particle moves in circle. Identify curvature nature. (2 Marks - [U/C, 1])
37	(i)	A fluid velocity field is given by $\vec{v}(t) = 3t\hat{i} + 2t^2\hat{j}$. Identify whether acceleration is constant. (3 Marks - [E/C, 2])
38	(ii)	Given $\vec{r}(t) = 3t\hat{i} + 2t^2\hat{j}$, identify the type and justify. (2 Marks - [E/C, 1])
39	(i)	An engineer assumes tangent and normal vectors are parallel. Identify and correct the error. (3 Marks - [U/C, 2])

40	(ii)	Identify role of unit tangent vector in motion analysis. (2 Marks - [U/C, 1])
41	(i)	An electric vehicle follows the trajectory $\vec{r}(t) = t^2\hat{i} + 3t\hat{j}$. Compute velocity and acceleration at $t = 2$. Analyze whether the motion is uniformly accelerated and justify. (5 Marks - [Ap/C, 3])
42	(i)	A pipe flow system is modeled by $\vec{F} = yz\hat{i} + xz\hat{j} + xy\hat{k}$. Compute divergence and analyze whether the system satisfies incompressibility. (5 Marks - [Ap/P, 3])
43	(i)	A drone path is $\vec{r}(t) = t^3\hat{i} + 2t^2\hat{j} + t\hat{k}$. Compute velocity, acceleration, and evaluate tangential component at $t = 1$. (5 Marks - [Ap/P, 3])
44	(i)	In an electromagnetic field, the vector field is $\vec{F} = 2x\hat{i} + 3y\hat{j}$. Compute divergence and evaluate whether the field conserves flux. Interpret physically. (5 Marks - [Ap/C, 3])
45	(i)	In an electrostatic field, potential $V(x, y, z) = xyz$. Compute gradient and evaluate directional derivative at $(1, 1, 1)$ along $\hat{i} + \hat{j} + \hat{k}$. Interpret physical meaning. (5 Marks - [Ap/P, 3])
46	(i)	In a fluid mixing system, velocity field is $\vec{F} = y\hat{i} - x\hat{j}$. Compute curl and analyze whether the flow is rotational. Interpret its impact on mixing efficiency. (5 Marks - [Ap/P, 3])
47	(i)	In an electric vehicle battery system, temperature distribution is modeled by $T(x, y, z) = x^2 + y^2 + z^2$. Compute the gradient and evaluate the direction of maximum temperature rise at point $(1, 1, 1)$. Interpret its physical significance for thermal management and justify. (5 Marks - [Ap/P, 3])
48	(i)	In an electromagnetic system, field $\vec{F} = 2x\hat{i} + 3y\hat{j} + 4z\hat{k}$. Compute curl and evaluate whether the field is conservative. Justify. (5 Marks - [Ap/P, 3])